Kenneth Gonzalez

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Education

CUNY Hunter College New York, NY
B.A. in Computer Science. *May 2027*
Relevant Coursework: *Operating System, Data Structures, Data Structures and Algorithms, Databases, Software Analysis and Design (I,II,III), Computer Architecture (I,II), Computer Theory*

Experience

Software Engineer Intern June 2026 – August 2026
Mastercard *New York, NY*
- Owned the Spring Boot 2.5 → 3.5 migration of Mastercard's primary checkout entry-point service processing 180M+ transactions daily across Click to Pay and Secure Card on File payment flows.
- Modernized 800+ lines across 30+ files, upgrading 40+ dependencies including Spring Security, Kafka, and Jakarta APIs while resolving breaking changes between internal Mastercard SDKs to enable Java 17+.
- Cut the 1,600+ functional test suite runtime from 21 to 18 minutes by consolidating Spring application contexts into a shared test configuration, then validated downstream-service compatibility with 0 regressions through PCF.

Software Engineer Intern June 2025 – August 2025
Mastercard *New York, NY*
- Reduced private-key discovery from ~3–5 minutes to seconds for 30+ Mastercard engineers by building a Spring Boot checkout-testing tool that automatically retrieved keys used to decrypt encrypted checkout payloads.
- Designed and implemented a private-key retrieval architecture for encrypted checkout payloads, migrating key lookups from a legacy repository to Mastercard's internal key service and automating decryption across test environments.
- Evaluated Redis against local caching and implemented an in-memory cache for private-key lookups, reducing repeated downstream service calls while avoiding unnecessary infrastructure, permissions, and operational complexity.

Software Engineer Intern May 2024 – August 2024
Intuit Credit Karma *Charlotte, NC*
- Shipped a production TypeScript/GraphQL API for an internal fraud tooling, aggregating all workflows and actions associated with a user into a single investigator view and serving 200 requests with 0 failures on launch day.
- Provisioned 4 New Relic dashboards with Terraform, authoring 9+ NRQL queries and 25 alerts across 20+ metrics to monitor production API latency, HTTP failures, and fraud-investigation traffic.
- Designed a technical design document with 2 senior engineers, defining relational schemas and 3 APIs to replace file-based action definitions and reduce a 3-day manual workflow-creation process.

Projects

Serverless Event Processing Pipeline | *YOLOv8, FastAPI, Python, React, Google Gemini, ElevenLabs API*
- Built an event-driven processing pipeline using AWS Lambda, SQS, DynamoDB, and S3, decoupling API ingestion from asynchronous processing to handle 10,000+ events during load testing.
- Implemented idempotent processing using DynamoDB conditional writes and SQS retry/DLQ policies, achieving zero duplicate writes across 500 intentionally duplicated events.
- Provisioned AWS Lambda, IAM, SQS, API Gateway, DynamoDB, S3, and CloudWatch infrastructure using Terraform and automated validation/deployment through GitHub Actions (CI/CD).

Mira | *YOLOv8, FastAPI, Python, React, Google Gemini, ElevenLabs API*
- Won 1st place among 70+ teams by building a real-time assistive web app that identifies people, vehicles, and obstacles and delivers spoken alerts for visually impaired users.

Skills

Languages: Java, Python, JavaScript/TypeScript, C++, HTML/CSS

Technologies: Spring Boot, React, Flask, Express.js, GraphQL, AWS, Terraform, Docker, Kubernetes

Databases/Tools: PostgreSQL, MongoDB, DynamoDB, RabbitMQ, Git/GitHub, Version Control, New Relic, Swagger, Linux

Concepts: Distributed Systems, Object Oriented Programming, Unit Testing, API Development, System Design, Microservices, JWT Authentication and Authorization, Hashing, Microservices, REST APIs, Caching, AI Powered Tools, Large Scale Web Application, System Monitoring, Live Deployment